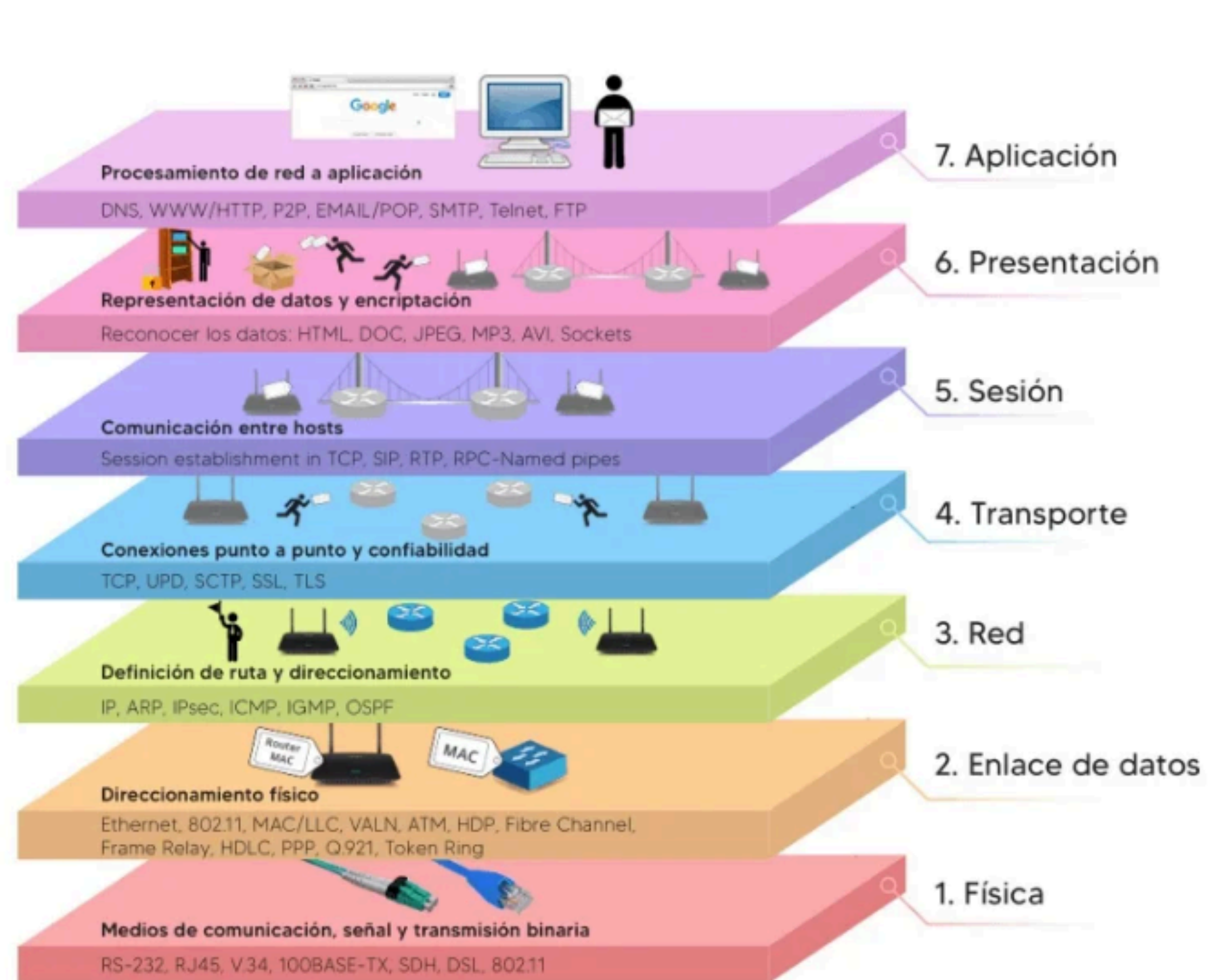


Redes y sus cosas



Capa OSI



MODELO OSI

7. APLICACIÓN

6. PRESENTACIÓN

5. SESIÓN

4. TRANSPORTE

3. RED

2. ENLACE DE DATOS

1. FÍSICA

MODELO TCP/IP

APLICACIÓN

TRANSPORTE

INTERNET

ACCESO
A LA RED

	Velocidad	Longitud Segmento	Tipo Cable
10BaseF	10 Mbps	2000 metros	Fibra óptica
10Base2	10 Mbps	185 metros	Cable coaxial
10Base5	10 Mbps	500 metros	Cable coaxial
10BaseT	10 Mbps	100 metros	Cable UTP
100BaseT4	100 Mbps	100 metros	Cable UTP
FastEthernet	100 Mbps	100 metros	Cable UTP y fibra óptica
GigabitEthernet	1000 Mbps	Cable: 25metros. Fibra monomodo: 2 Kilometros. Fibra multimodo: 500 metros	Fibra óptica y cable UTP
10GBaseT	1000 Mbps	100 metros	Cable de par trenzado
1000BaseSX	1000 Mbps	550 metros	Fibra óptica multimodo
1000BaseLX	1000 Mbps	5000 metros	Fibra óptica monomodo



RJ-45



RJ-11



RJ-21 (25-pair)



DE-9 (Female)



DB-25 (Male)



DB-60 (Male)

Transceptores

GBICs



1000Base-SX/LX



1000Base-T



Cisco GBIC



1000Base-SX/LX SFP



1000Base-T SFP



X2 (10Gig)

Optical Terminations



ST (Straight Tip)



SC (Subscriber Connector)

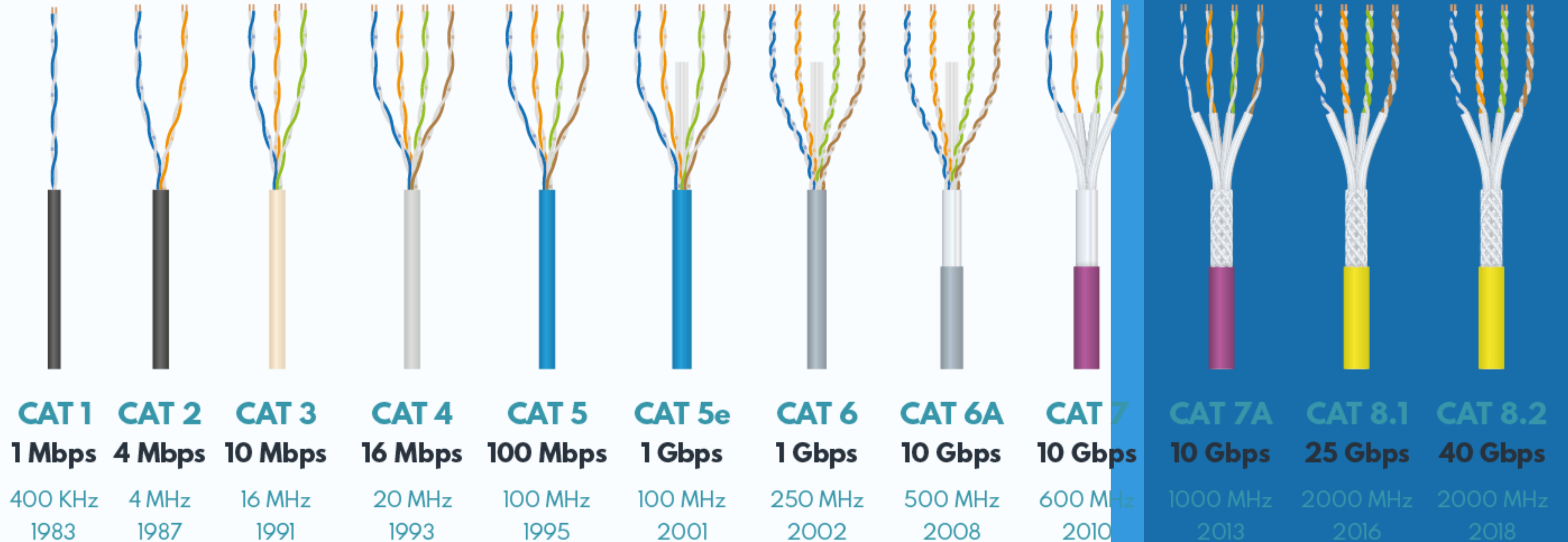


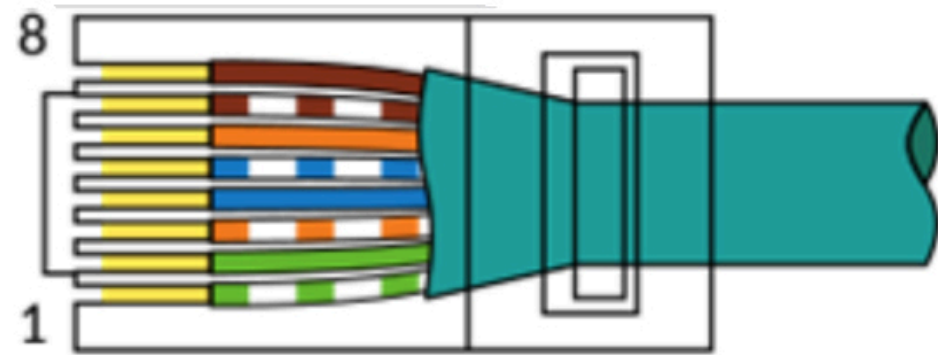
LC (Local Connector)



MT-RJ

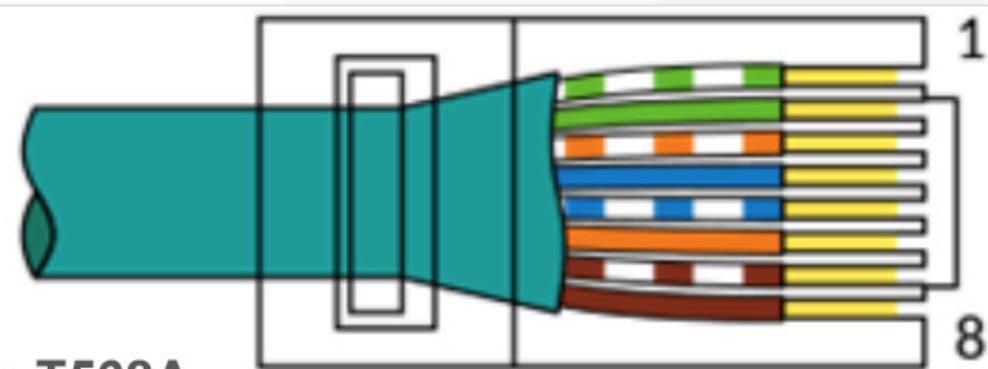
History of Ethernet LAN Cables' Categories



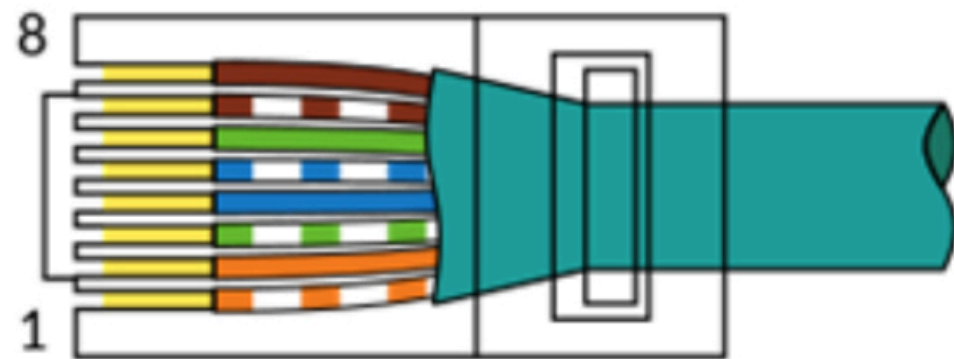


T568A

Latiguillo conexión en línea T568A

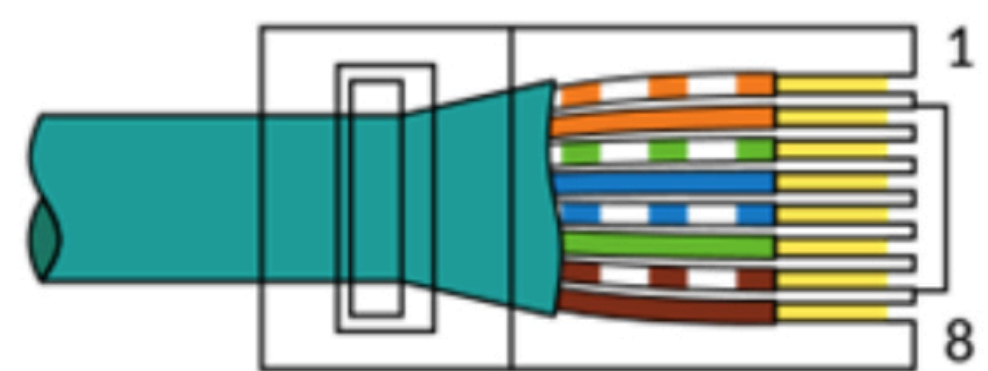


T568A



T568B

Latiguillo conexión en línea T568B



T568B



tutorialesenlinea.es

LOS CONECTORES DE LA FIBRA OPTICA

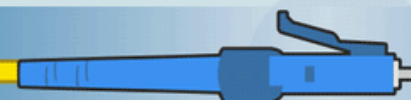
CONECTORES, PULIDOS Y COLORES: GUÍA SENCILLA



FC FERRULE CONNECTOR
CONECTOR DE FERRULE



ST STRAIGHT TIP
PUNTA RECTA



LC LUCENT CONNECTOR
CONECTOR LUCENT



SC SUSCRIPTOR CONNECTOR
CONECTOR DE SUSCRIPTOR

TIPOS DE PULIDO

PC



PHYSICAL
CONTACT

UPC



ULTRA
PHYSICAL
CONTACT

APC



ANGLED
PHYSICAL
CONTACT

COLOR DEL CONECTOR

RECUERDA LA NOMENCLATURA DE LOS CONECTORES

CONECTOR

SC/PC

PULIDO



FIBRA 62.5/125 μm



FIBRA 50/125 μm



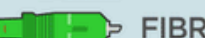
FIBRA 50/125 μm LASER OPTIMIZED



FIBRA OM5



FIBRA MONOMODO



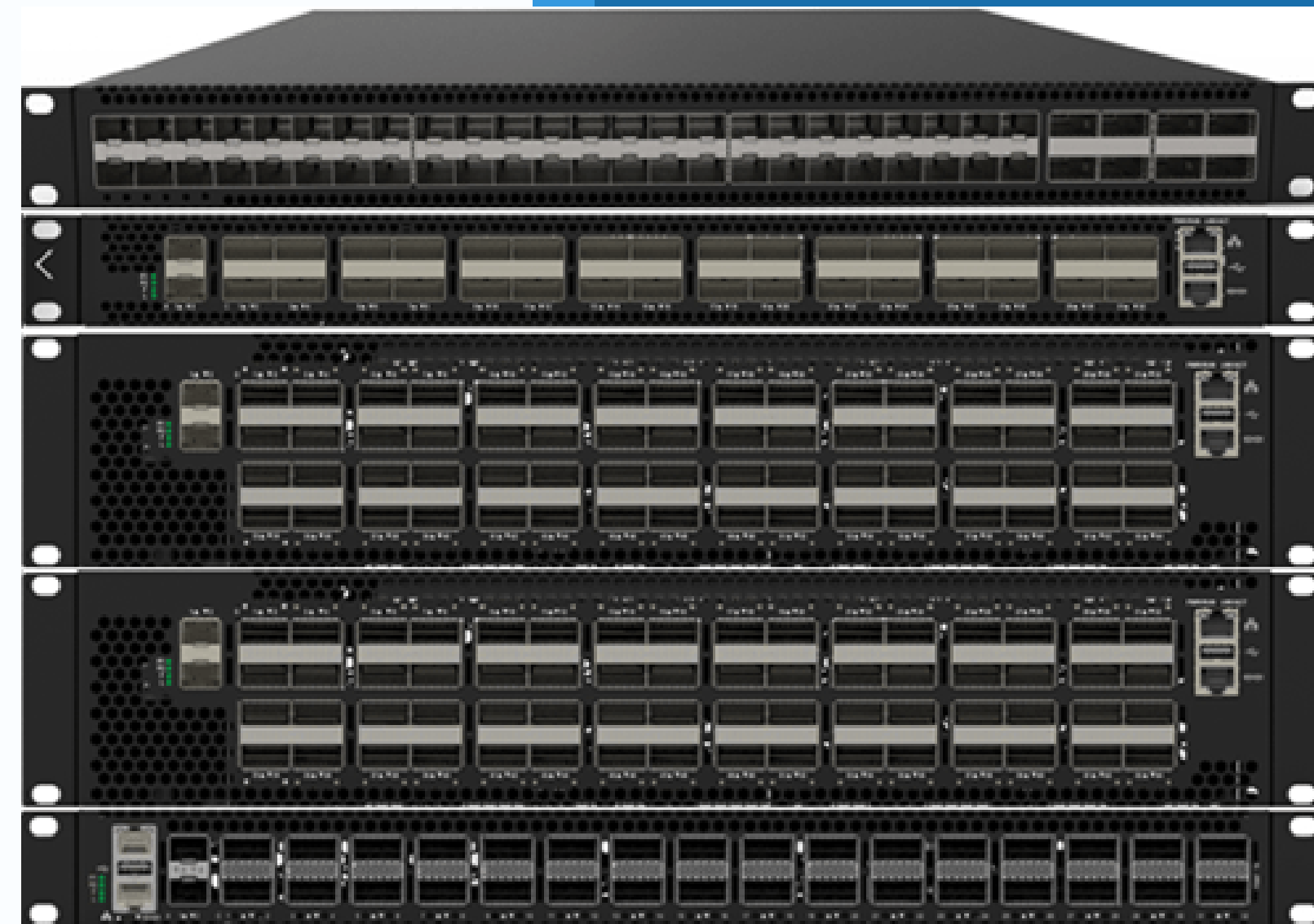
FIBRA MONOMODO APC

PROMAX
www.promax.es





SWITCH



Estados de puertos

BEAR

01

Bloqueado (Blocked) • Al iniciar el switch •

02

Escuchando (Listening) • Esperando mensajes STP para asegurarse de que no hay bucles

03

Aprendiendo (Learning) • Recibiendo tramas y guardando direcciones MAC en la tabla •

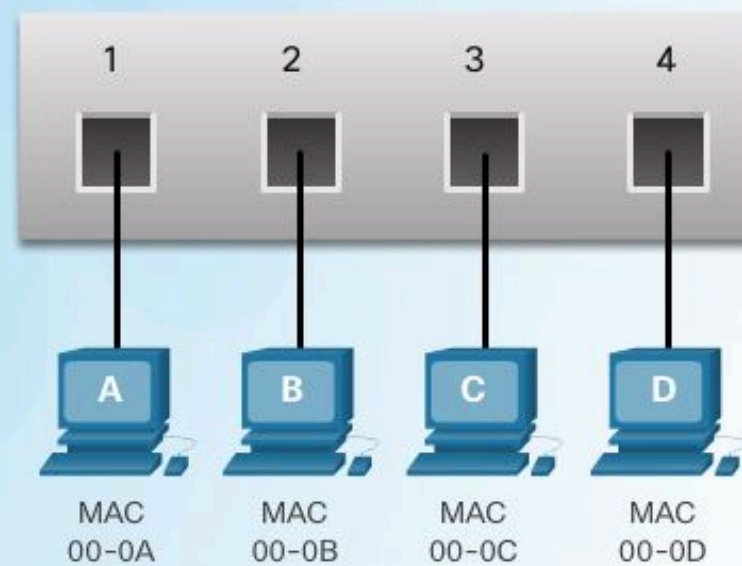
04

Reenviando (Forwarding)

Tabla de direcciones MAC

Aprendizaje: Examinar la dirección MAC de origen

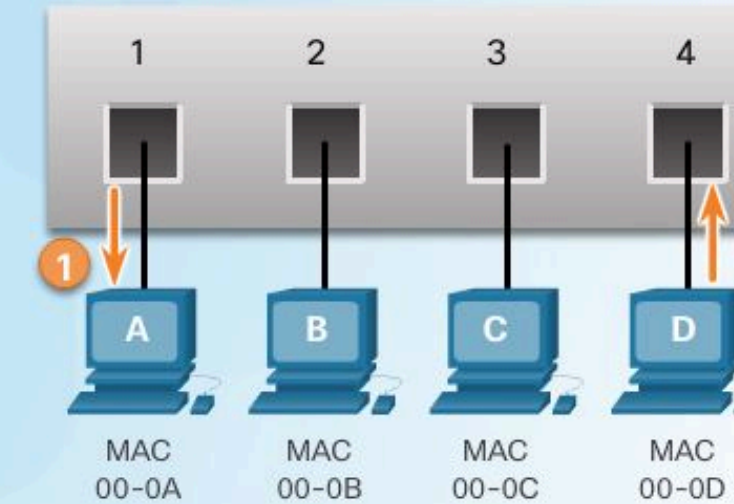
Tabla de direcciones MAC	
Puerto	Dirección MAC



Las direcciones MAC están acortadas con fines de demostración.

Reenvío: Examinar la dirección MAC de destino

Tabla de direcciones MAC	
Puerto	Dirección MAC
1	00-0A
4	00-0D



MAC de destino 00-0A	MAC de origen 00-0D	Tipo	Datos	FCS
-------------------------	------------------------	------	-------	-----

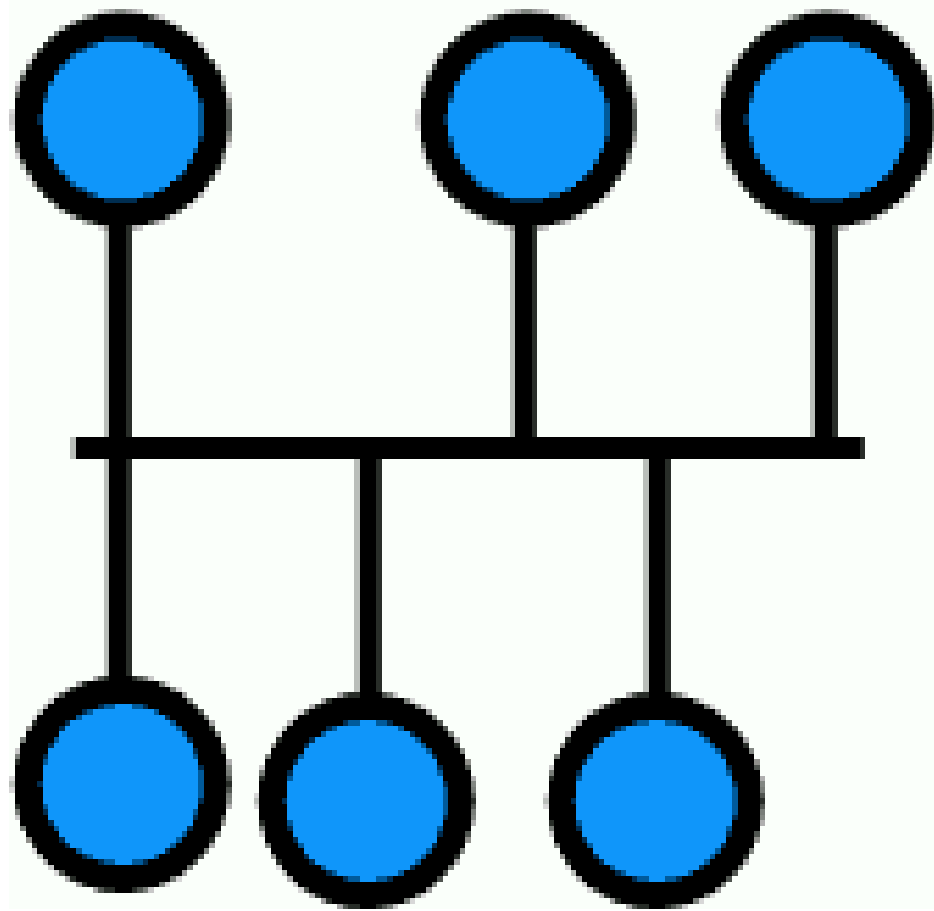
Las direcciones MAC están acortadas con fines de demostración.


```
switch1#show mac address-table
```

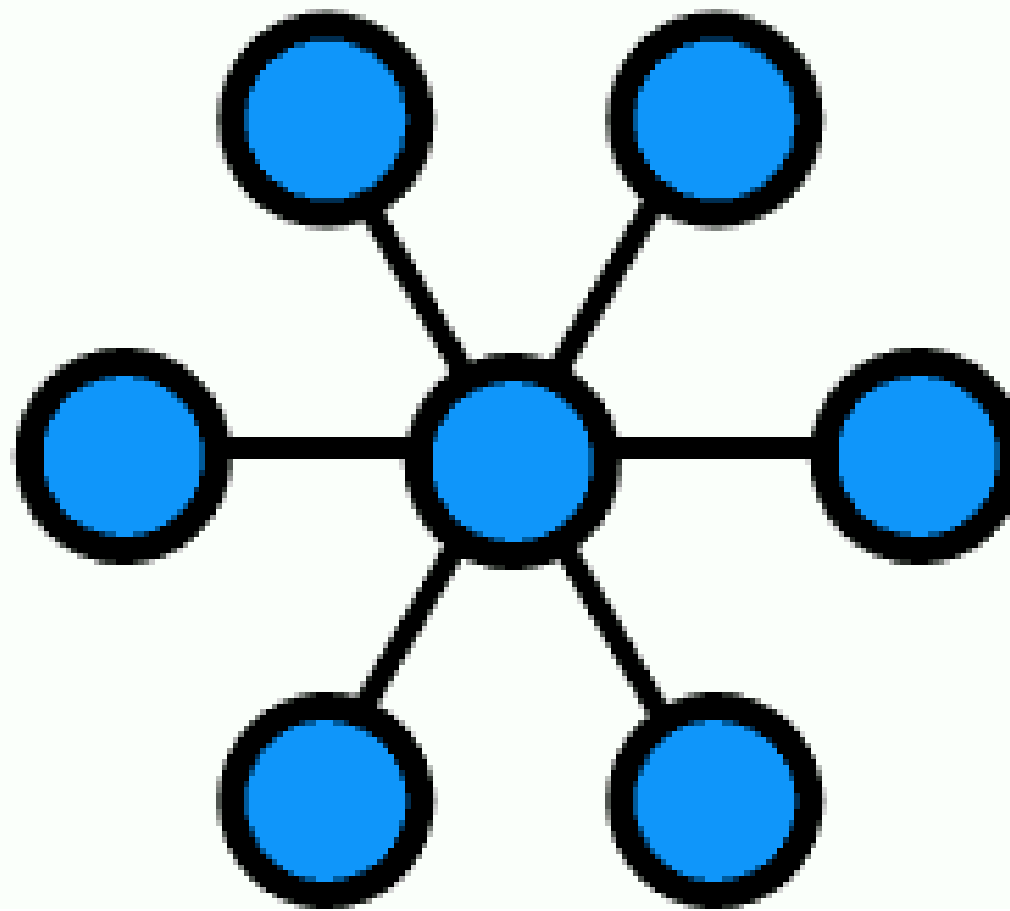
Mac Address Table

Vlan	Mac Address	Type	Ports
All	0011.5ccc.5c00	STATIC	CPU
All	0100.0ccc.cccc	STATIC	CPU
All	0100.0ccc.cccd	STATIC	CPU
All	0100.0cdd.dddd	STATIC	CPU
1	0009.5b44.9d2c	DYNAMIC	Fa0/1
1	000f.66e3.352b	DYNAMIC	Fa0/1
1	0012.8015.c940	DYNAMIC	Fa0/24
1	0012.8015.c941	DYNAMIC	Fa0/24
1	001a.adb3.bef7	DYNAMIC	Fa0/1
1	0025.2266.d104	DYNAMIC	Fa0/1
1	0026.b865.313e	DYNAMIC	Fa0/1
1	64a7.6973.8e4d	DYNAMIC	Fa0/1
1	6c71.d976.fce7	DYNAMIC	Fa0/1
1	74f6.12d4.1e1c	DYNAMIC	Fa0/1
1	a477.3344.98b6	DYNAMIC	Fa0/1

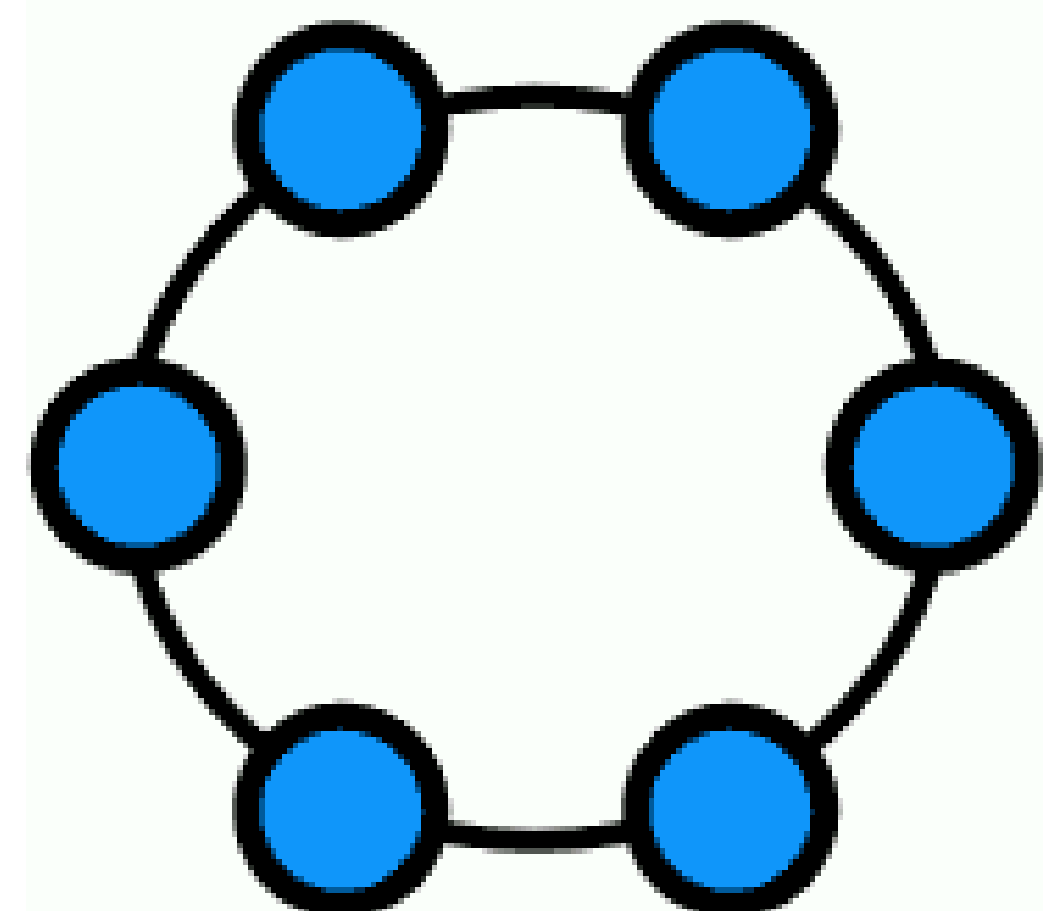
Bus



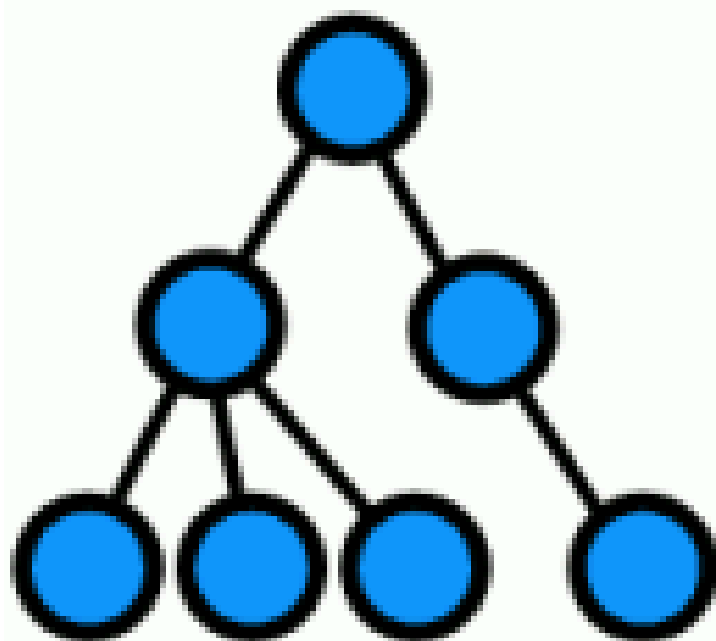
Estrella



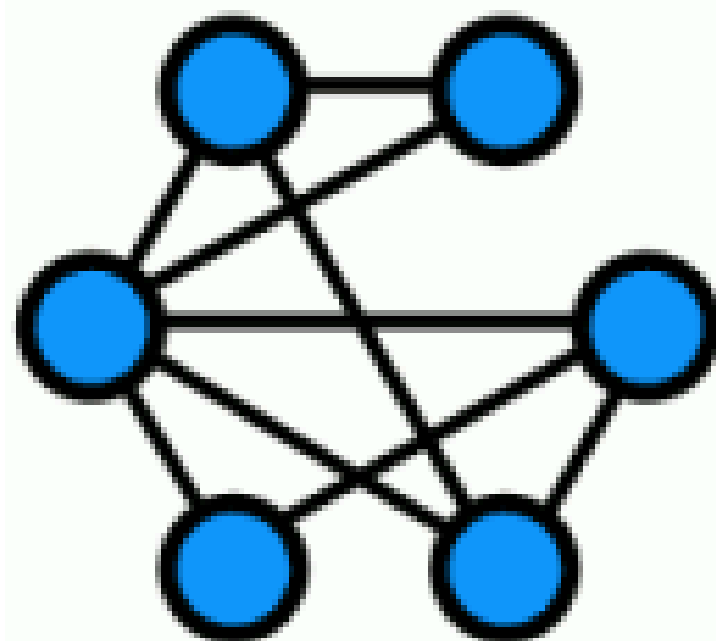
Anillo



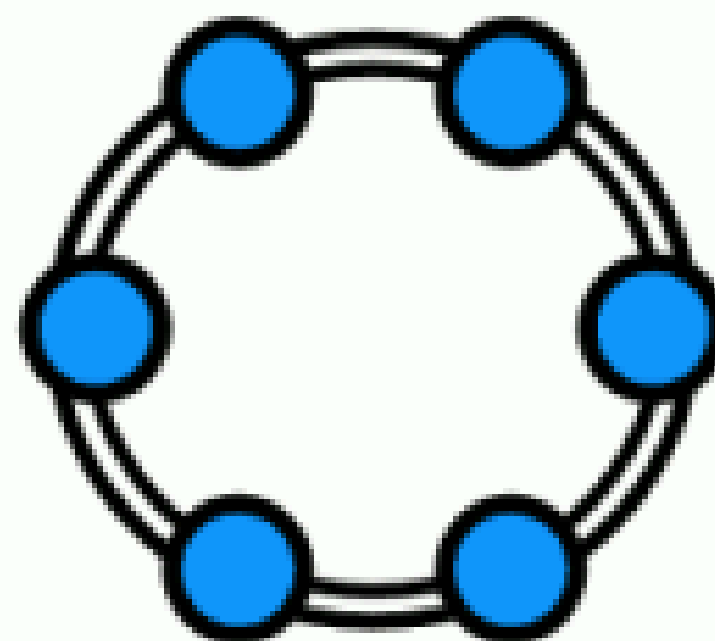
Árbol



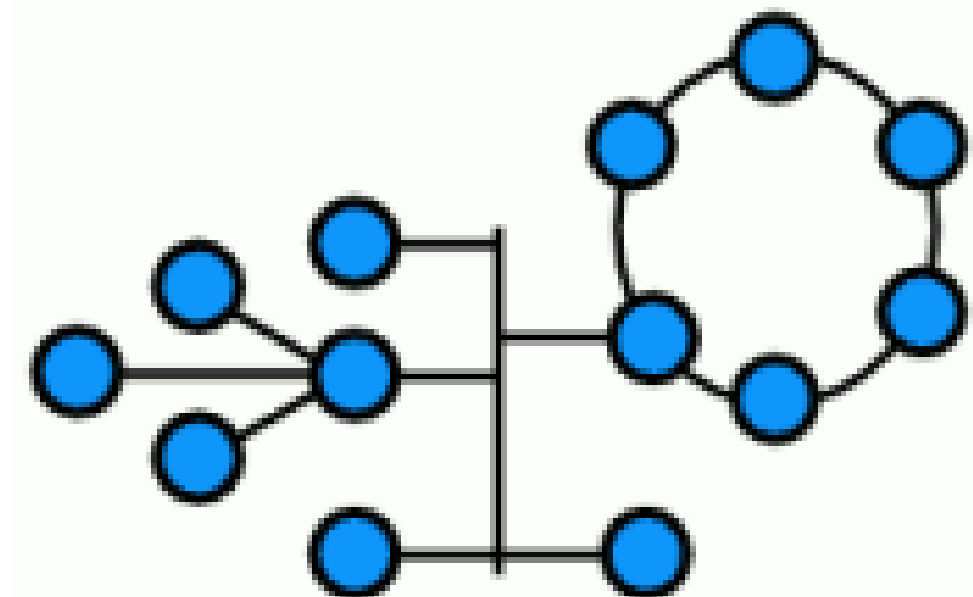
Malla



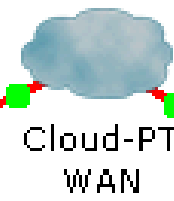
Doble Anillo



Mixta



Norte



Cloud-PT
WAN

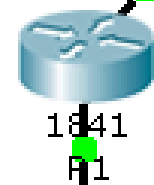
Core



2811
C1



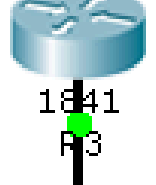
2811
C2



1841
R1

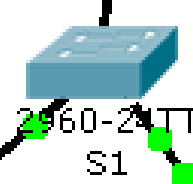


2621XM
R2

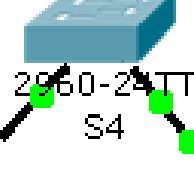


1841
R3

Distribution



2960-24TT
S1



2960-24TT
S4



2960-24TT
S7

Oeste

Access



2960-24TT
S2



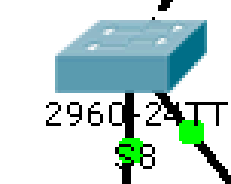
2960-24TT
S3



2960-24TT
S5



2960-24TT
S6



2960-24TT
S8



2960-24TT
S9



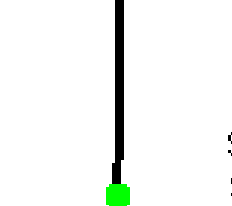
7960
Teléfono1



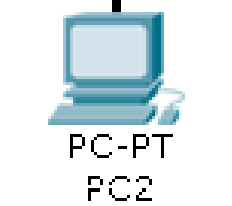
Printer-PT
Impresora1



PC-PT
PC1



Server-PT
Servidor1



PC-PT
PC2



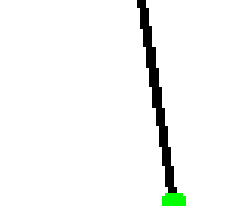
7960
Teléfono2



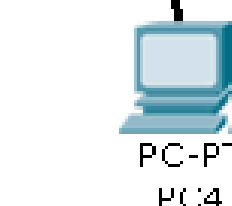
Printer-PT
Impresora2



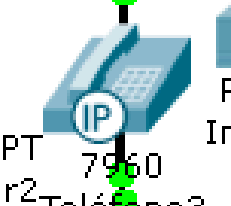
PC-PT
PC3



Server-PT
Servidor2



PC-PT
PC4



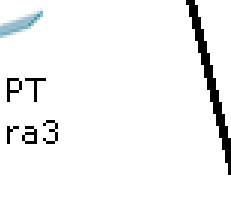
7960
Teléfono3



Printer-PT
Impresora3



PC-PT
PC5



Server-PT
Servidor3



PC-PT
PC6

SUR

Este



Característica	Gateway	Router
Función principal	Conectar redes con diferentes protocolos y arquitecturas.	Direccionar paquetes de datos dentro de o entre redes similares.
Protocolos	Traduce y comunica entre diversos protocolos.	Usa protocolos de enrutamiento para determinar las rutas.
Aplicación	Entre redes significativamente diferentes.	Mayormente dentro de redes compatibles o ligeramente distintas.
Uso	Esencial en entornos donde las redes a conectar son incompatibles.	Crucial para la construcción y gestión de redes y subredes complejas.

Tabla de routing IPv4 del R1



```
R1# show ip route
```

```
Gateway of last resort is not set
```

```
10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
D    10.1.1.0/24 [90/2170112] via 209.165.200.226, 00:00:05,
    Serial0/0/0
D    10.1.2.0/24 [90/2170112] via 209.165.200.226, 00:00:05,
    Serial0/0/0
192.168.10.0/24 is variably subnetted, 2 subnets, 3 masks
C    192.168.10.0/24 is directly connected, GigabitEthernet0/0
L    192.168.10.1/32 is directly connected, GigabitEthernet0/0
L    192.168.11.1/32 is directly connected, GigabitEthernet0/1
209.165.200.0/24 is variably subnetted, 2 subnets, 3 masks
C    209.165.200.224/30 is directly connected, Serial0/0/0
L    209.165.200.225/32 is directly connected, Serial0/0/0
```


Thank you

○ SD-WAN

